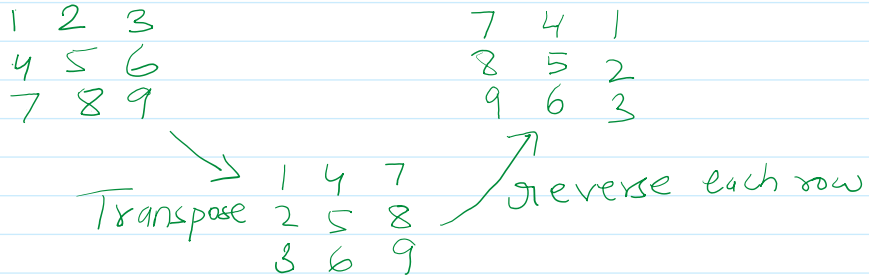
Doubts on Array Programs

10 October 2026 07:49

Rotate Matrix 90° Rotate a square matrix clockwise by 90°. Extend the program to rotate by 180° and 270°.



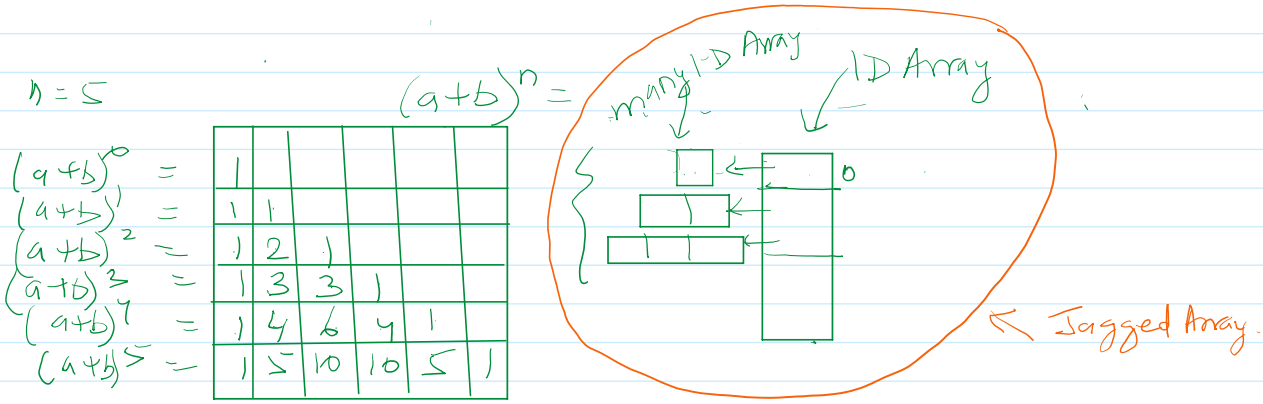
Pascal's Triangle Generate Pascal's Triangle using a 2D array. Print it in triangular format.

$(a+b)^0 = 1$

$(a+b)^2 = 1a^2 + 2ab + 1b^2$

$= 1a^2 \cdot b^0 + 2a^1b^1 + 1 \cdot a^0 b^2$

$(a+b)^3 = 1a^3 \cdot b^0 + 3a^2b^1 + 3a^1b^2 + 1a^0b^3$



1. **Pascal's Triangle** Generate Pascal's Triangle using a 2D array. Print it in triangular format.

2. **Sudoku Validator** Input a 9x9 Sudoku grid. Validate if:
   - Each row contains digits 1–9 without repetition.
   - Each column contains digits 1–9 without repetition.
   - Each 3x3 subgrid contains digits 1–9 without repetition.

3. **Image Pixel Manipulation** Represent a grayscale image as a 2D array (values 0–255). Implement:
   - Brightness increase/decrease.
   - Horizontal flip.
   - Vertical flip.
   - Inversion (255 – value).

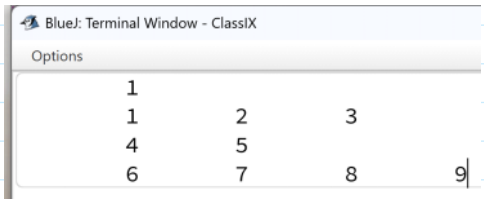
Jagged Arrays

```
package MyArrays.TwoDimensionalArray;
import java.util.*;


/**
 * Write a description of class JaggedArray here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class JaggedArray
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int[][] a = {{1}, {1, 2, 3}, {4, 5},{6,7,8,9}};
        int i, j;
        for(i = 0; i < a.length; i++){
            for(j = 0; j < a[i].length; j++){
                System.out.print("\t"+a[i][j]);
            }
            System.out.print("\n");
        }
    }
}
```

Jagged array

Output:



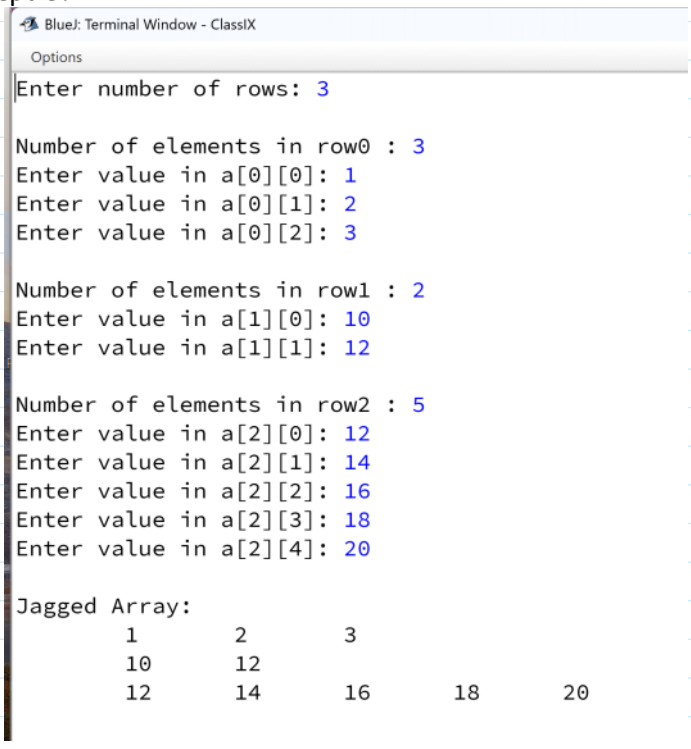

```
package MyArrays.TwoDimensionalArray;
import java.util.*;


/**
 * Write a description of class JaggedArray here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class JaggedArray
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        // int[][] a = {{1}, {1, 2, 3}, {4, 5},{6,7,8,9}};
        System.out.print("\fEnter number of rows: ");
        int rows = sc.nextInt();
```

```
        int[][] a = new int[rows][];
        int i, j;

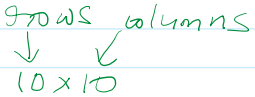
        //feedind data in jagged array
        for(i = 0; i < a.length; i++){
            System.out.print("\nNumber of elements in row" +i+" : ");
            int cols =  sc.nextInt();
            a[i] = new int[cols]; //i+1 instead of columns
            for(j = 0; j < a[i].length; j++){
                System.out.print("Enter value in a["+i+"]["+j+"]: ");
                a[i][j] = sc.nextInt();
            }
        }
        System.out.print("\nJagged Array: \n");
        for(i = 0; i < a.length; i++){
            for(j = 0; j < a[i].length; j++){
                System.out.print("\t"+a[i][j]);
            }
            System.out.print("\n");
        }
    }
}
```

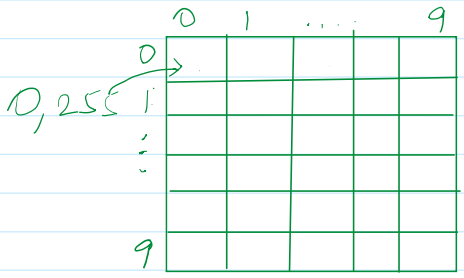
Output:



4. **Image Pixel Manipulation** Represent a grayscale image as a 2D array (values 0–255). Implement:
   - Brightness increase/decrease.
   - Horizontal flip.
   - Vertical flip.
   - Inversion (255 – value).







10 x 10

0 ← black
255 ← white
Values in between represents shades of gray.

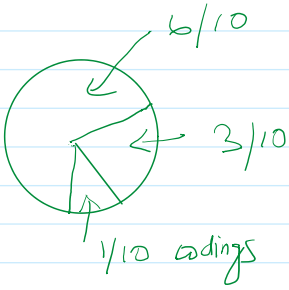
1. Brightness adjustment:-

newPixel = oldPixel + amount

The result must be clamped to the range 0-255. For example, increasing 240 by 30 produces 255 not 270.

2. Horizontal flip:- The program swaps the leftmost and rightmost pixels in each row; then moves inward.

original row: 10 20 30 40 50
Flipped: 50 40 30 20 10



3. Vertical flip: The program swaps the first row with last row, then moves inward.

original rows:

10	20	30
40	50	60
70	80	90
5	15	25
35	45	55

Flipped:

35	45	55
5	15	25
70	80	90
40	50	60
10	20	30

Inversion:-

newPixel = 255 - oldPixel

For example, 0 becomes 255, 255, becomes 0, & 100 becomes 155.